Appendix 2. Matrices, matrix operations, and determinants

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A2.1 Matrices and equality

An $m \times n$ matrix is a rectangular array with m rows and n columns,

$$A = (a_{ij})_{1 \le i \le m, 1 \le j \le n}.$$

It is square when $m = n$. Two matrices are equal iff dimensions match and every corresponding entry agrees. Similar-looking entries with different dimensions do not establish equality. A transpose exchanges row and column indices and dimensions.

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A2.2 Addition and scalar multiplication

For equal dimensions, define $(A + B)_{ij} = a_{ij} + b_{ij}$. For real λ, define $(\lambda A)_{ij} = \lambda a_{ij}$. Entrywise arithmetic gives commutativity and associativity of addition and the distributive laws.

A 2×3 matrix cannot be added to a 3×4 matrix; missing dimensions cannot be padded with zeros. Zero matrices also require matching dimensions, so $0_{2\times 3} + 0_{3\times 2}$ is undefined.

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A2.3 Matrix multiplication

For A of size $m \times n$ and B of size $n \times p$, their product has size $m \times p$, with

$$(AB)_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}.$$

Column j of the product is A acting on column j of B. Row i is a linear combination of B's rows with coefficients from A's row i.

For general sizes $m \times n$ and $p \times q$, AB requires $n = p$, and BA requires $q = m$. Both can exist without being equal. To compare them as same-sized arrays requires $m = n$, hence square factors under these simultaneous dimension conditions. Commutativity is not a prerequisite for multiplication and does not generally hold.

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A2.4 Associativity, distributivity, and zero divisors

For compatible dimensions, $(AB)C = A(BC)$ and $A(B + C) = AB + AC$. Entrywise,

$$((AB)C)_{ij} = \sum_l \left(\sum_k a_{ik} b_{kl} \right) c_{lj} = \sum_k a_{ik} \left(\sum_l b_{kl} c_{lj} \right) = (A(BC))_{ij}.$$

Finite-sum distribution and rearrangement justify the steps. The matrix ring $M_n(F)$ is noncommutative for $n \geq 2$ and has zero divisors. For example,

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = 0$$

with both factors nonzero. Thus a zero product need not have a zero factor.

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A2.5 Transpose, identity, and inverse

Define $(A^T)_{ij} = a_{ji}$; then $(AB)^T = B^T A^T$. The identity I_n has diagonal ones and other entries zero, and acts as identity whenever dimensions are compatible.

For square A, a matrix B satisfying $AB = BA = I_n$ is its unique inverse, written A^{-1}. In size two,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}, \qquad ad - bc \neq 0.$$

Rectangular left or right inverses are separate concepts and are not denoted by an ordinary two-sided A^{-1}.

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A2.6 The determinant's domain and definition

For a square $n \times n$ matrix,

$$\det A = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^{n} a_{i,\sigma(i)}.$$

There are $n!$ terms. The determinant is multilinear and alternating in rows, meaning linear in one row with others fixed and sign-changing under row exchange. It does not change sign under arbitrary exchanges of individual entries. For size two it is $ad - bc$.

A nonsquare matrix has no determinant; choosing an arbitrary square block does not define one for the whole matrix. Cofactor expansion is a computational theorem on the same square domain.

A2.7 Minors, cofactors, and expansion

Let M_{ij} be the determinant after deleting row i and column j. Define $C_{ij} = (-1)^{i+j} M_{ij}$. Along any row,

$$\det A = \sum_{j=1}^{n} a_{ij} C_{ij}.$$

A column version also holds. The alternating signs form a checkerboard; missing them is a common error. Expansion gives recursive computation, while elimination is generally preferable for large sizes.

A2.8 Determinant properties and invertibility

For same-sized square matrices,

$$\det(AB) = \det A \det B, \qquad \det(A^T) = \det A.$$

Swapping rows negates the determinant; scaling one row by c multiplies it by c; adding a multiple of one row to another leaves it unchanged. Dependent or repeated rows give zero.

The inverse exists iff $\det A \neq 0$. Then $AX = B$ has unique solution $X = A^{-1}B$, and Cramer's rule also divides by $\det A$. A zero determinant may give no solutions or infinitely many over reals. Invertibility is a criterion; computing the inverse still costs work.

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A2.9 Boundary checks

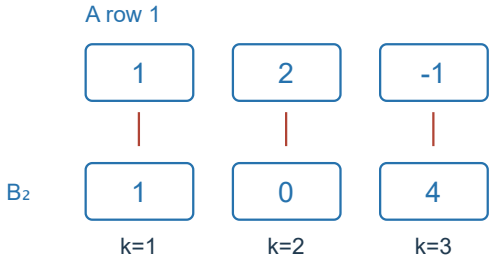
Addition needs identical dimensions; scalar multiplication acts on any fixed-size matrix. Multiplication requires $A_{m\times n}B_{n\times p}$. Transposing a product reverses order. Determinants and ordinary inverses require square matrices, and inverses require nonzero determinant. The formula $A^{-1}B$ for a unique system solution requires these conditions on the coefficient matrix.

State dimensions before numerical calculations. An undefined operation is not a zero result. Group calculations, coding matrices, and signal transformations need the same checks.

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A2.10 One contracted index produces one entry

Matrix inner index k: match one row and one column by position



$(AB)_{12}$ = 1×1 + 2×0 + (−1)×4

= 1 + 0 − 4 = −3

All entries: AB = [5 −3 ; 1 8]

(2×3)(3×2) → 2×2; sum over k.

Red lines match the same k; row and column lengths must agree.

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 3 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 1 \\ 1 & 0 \\ -1 & 4 \end{pmatrix}, \quad AB = \begin{pmatrix} 5 & -3 \\ 1 & 8 \end{pmatrix}.$$

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A2.10 One contracted index produces one entry (continued)

Entries are $2+2+1=5$, $1+0-4=-3$, $0+3-2=1$, and $0+0+8=8$. Each product term shares inner index k; this is not entrywise multiplication. The product BA exists but is 3×3, unlike AB's 2×2.

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A2.11 Alternating multilinearity from the definition

Entries here are real; the proof works over any field. Each Leibniz term uses exactly one entry from a specified row, giving linearity in that row. Exchange rows r, s and pair permutations by exchanging the selected columns in those rows. Their signs reverse. If rows agree, each pair has identical products and opposite signs, summing to zero. This argument never divides by two, so remains valid in characteristic two; merely writing $\det A = -\det A$ would not suffice there.

Expand each row $v_i = \sum_j a_{ij} e_j$ by multilinearity. Repeated basis rows vanish by alternation. Remaining terms permute the basis and have value $\operatorname{sgn}(\sigma)$ times the standard-basis value. Thus any alternating multilinear function with $d(I) = 1$ must be

$$d(A) = \sum_{\sigma} \operatorname{sgn}(\sigma) \prod_{i} a_{i,\sigma(i)} = \det A.$$

A row expressed as a combination of others gives repeated rows in every expansion term, hence zero. This also proves invariance under adding another row's multiple.

A2.12 Product and transpose theorems

Row i of AB is $\sum_k a_{ik} \operatorname{row}_k(B)$. Multilinearity gives

$$\det(AB) = \sum_{k_1,\ldots,k_n} \left(\prod_i a_{ik_i} \right) \det(B_{k_1,*}, \ldots, B_{k_n,*}).$$

Repeated k's vanish. Distinct choices form a permutation $k_i = \sigma(i)$, whose reordered rows have determinant $\operatorname{sgn}(\sigma) \det B$. Therefore

$$\det(AB) = \left(\sum_{\sigma} \operatorname{sgn}(\sigma) \prod_i a_{i,\sigma(i)} \right) \det B = \det A \det B.$$

This proves the theorem for arbitrary size without assuming either matrix invertible.

Replace σ by σ^{-1} in Leibniz, with unchanged sign, and rearrange products to prove $\det(A^T) = \det A$. Row properties therefore hold for columns too. For a triangular matrix, every nonidentity permutation selects a structural zero, leaving the diagonal product.

Separately,

A2.12 Product and transpose theorems (continued)

$$(AB)_{ij}^T = \sum_k a_{jk} b_{ki} = \sum_k (B^T)_{ik} (A^T)_{kj},$$

proving equality of transpose matrices. Equality of determinants alone is a different claim.

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A2.13 Cofactors, adjugate, and inverse

Group Leibniz terms according to column j selected in fixed row i. Deleting that row and column changes the remaining permutation sign by $(-1)^{i+j}$, proving row expansion. Transpose gives column expansion. For $n = 1$, use $\det(0 \times 0) = 1$ as the recursion base.

Define $\operatorname{adj}(A)_{ji} = C_{ij}$, reversing indices. Then

$$(A \operatorname{adj} A)_{ij} = \sum_k a_{ik} C_{jk}.$$

For $i = j$ this expands $\det A$. Otherwise it expands along row j after replacing that row by row i, giving two equal rows and determinant zero. Thus $A \operatorname{adj} A = (\det A) I$. Column expansion similarly gives $(\operatorname{adj} A) A = (\det A) I$.

If $\det A \neq 0$, dividing constructs the two-sided inverse. Conversely, an inverse and the product theorem give $\det A \det(A^{-1}) = 1$, so the determinant is nonzero. If B, C are inverses, $B = B(AC) = (BA)C = C$. Likewise $(AB)^{-1} = B^{-1} A^{-1}$, with reversed order.

A2.14 Full elimination of a third-order determinant

$$\begin{pmatrix} 2 & 1 & 0 \\ 0 & 3 & 1 \\ 1 & -1 & 2 \end{pmatrix} \xrightarrow{R_3 - \frac{1}{2}R_1} \begin{pmatrix} 2 & 1 & 0 \\ 0 & 3 & 1 \\ 0 & -\frac{3}{2} & 2 \end{pmatrix} \xrightarrow{R_3 + \frac{1}{2}R_2} \begin{pmatrix} 2 & 1 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & \frac{5}{2} \end{pmatrix}.$$

Both operations preserve the determinant. The triangular result gives $2 \cdot 3 \cdot (5/2) = 15$. Expanding the original first row gives $2(6+1) - 1(-1) = 15$, agreeing.

A zero pivot may require swapping with a nonzero one, recording a minus sign for each swap; never divide by zero. A zero unprocessed column offers no pivot. With all n pivots, the triangular diagonal product is nonzero. Fewer pivots give a zero echelon row and determinant zero. Elimination uses $O(n^3)$ arithmetic operations instead of $n!$ explicit terms; exact rational bit complexity is a separate question.

A2.15 Inversion solves all identity columns together

Apply identical row operations to $[A|I]$:

$$\left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & 3 & 0 & 1 \end{array}\right] \xrightarrow{R_1/2} \left[\begin{array}{cc|cc} 1 & 1/2 & 1/2 & 0 \\ 1 & 3 & 0 & 1 \end{array}\right] \xrightarrow{R_2 - R_1} \left[\begin{array}{cc|cc} 1 & 1/2 & 1/2 & 0 \\ 0 & 5/2 & -1/2 & 1 \end{array}\right].$$

Scale R_2 by $2/5$, then use $R_1 - R_2/2$:

$$\left[\begin{array}{cc|cc} 1 & 0 & 3/5 & -1/5 \\ 0 & 1 & -1/5 & 2/5 \end{array}\right].$$

Each row operation is invertible left multiplication by an elementary matrix. Their product E satisfies $EA = I$. Since $\det A = 5 \neq 0$, $E = A^{-1}$. For $b = (5, 7)^T$, the solution is $((15-7)/5, (-5+14)/5) = (8/5, 9/5)$, which satisfies both $2x + y = 5$ and $x + 3y = 7$.

A2.16 Cramer and singular-system branches

If A's columns are $a_1, \ldots, a_n$ and $b = \sum_j x_j a_j$, replace column i by b. Expand linearly. Every $j \neq i$ term repeats another column and vanishes; the $j = i$ term is $x_i \det A$. Hence $x_i = \det A_i(b) / \det A$, assuming nonzero determinant and the unique solution.

For zero determinant, inspect the augmented system:

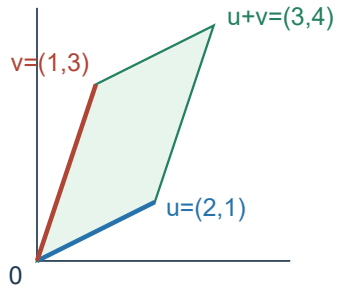
$$\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} x = \begin{pmatrix} 3 \\ 6 \end{pmatrix} \Rightarrow x = (3 - 2t, t),\ t \in \mathbb{R}.$$

With the same matrix and right side $(3, 7)^T$, subtract twice the first row from the second to get $0 = 1$, so no solution exists. A zero determinant alone does not select the branch.

Generally an echelon row $[0 \cdots 0 | c]$ with $c \neq 0$ makes a system inconsistent. Otherwise choose free variables arbitrarily and uniquely back-substitute pivot variables. At least one free variable gives infinitely many solutions over reals, but only finitely many over a finite field.

A2.17 Geometric and algebraic determinant values

A columns u=(2,1)、v=(1,3)：signed area det(A)=5



Counterclockwise: 0 → u → u+v → v.

det(A)=2×3−1×1=5 > 0

Swap columns: det=−5; area remains 5.

Dependent columns collapse to a segment; det=area=0.

For columns $u = (2, 1), v = (1, 3)$, the unit square maps to vertices $0, u, u + v, v$. The determinant $6 - 1 = 5$ preserves orientation and multiplies area by five. Swapping columns reverses orientation and gives minus five with the same area. Dependent columns collapse the image to a segment of area zero.

A2.17 Geometric and algebraic determinant values (continued)

In two dimensions, signed area is $u_1v_2 - u_2v_1$, derivable by base times height or subtraction of triangles, matching Leibniz. Ordinary area is its absolute value. Three-dimensional volume also uses absolute determinant; an abstract field need not have a real area interpretation, while the algebraic definition remains valid.

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References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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